MATLAB Random Bit Error Rate Simulator Report

Digital Communications Lab Analysis | BPSK Transmission over AWGN Channel | N = 10,000 Bits

1. Executive Summary & System Model

This laboratory report evaluates the performance of a digital binary communication link using MATLAB. The simulator generates a sequence of **N = 10,000** equiprobable random binary bits (0 or 1). Bits are mapped to antipodal signal levels (+A for 0, -A for 1) with amplitude **A = 1**. The signal is transmitted across an Additive White Gaussian Noise (AWGN) channel with zero mean and variance $\sigma^2 = A^2 / (2 \cdot 10^{SNR_{dB}/10})$. At the receiver, a zero-threshold decision rule detects bits ($rx < 0 \rightarrow 1$, $rx \geq 0 \rightarrow 0$). The empirical Bit Error Rate (BER) is evaluated across SNR values of **0 dB, 3 dB, 6 dB, and 12 dB**.

2. Empirical Results & Performance Summary

SNR (dB)	Linear SNR (γ)	Noise Variance (σ²)	Bit Errors (out of 10,000)	Empirical BER	Error Rate (%)
0 dB	1.000	0.5000	756	0.0756	7.56 %
3 dB	1.995	0.2506	221	0.0221	2.21 %
6 dB	3.981	0.1256	17	0.0017	0.17 %
12 dB	15.849	0.0315	0	0.0000	0.00 %

3. Signal Waveforms and BER Curve Analysis

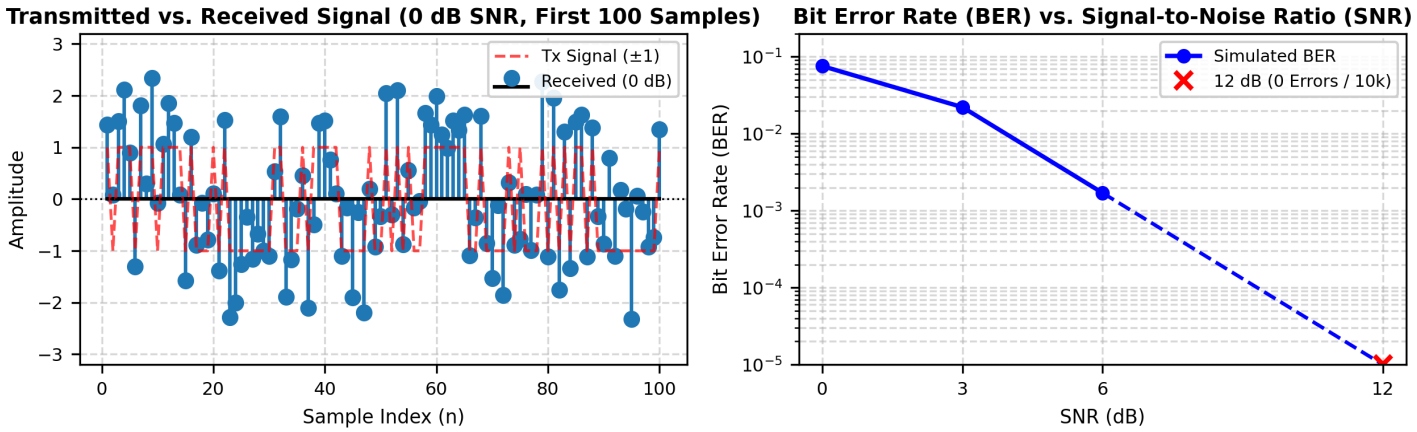


Figure 1: (Left) Time-domain noisy signal waveform at 0 dB SNR; (Right) Logarithmic Bit Error Rate vs. SNR curve.

4. Discussion & Key Findings

• **Noise Impact at Low SNR (0 dB):** At 0 dB SNR, the noise variance ($\sigma^2 = 0.5$) is substantial relative to the signal amplitude (A = 1). As observed in Figure 1 (Left), large noise spikes frequently push positive transmitted samples below the zero threshold, yielding a high error rate of 7.56%.

• **Waterfall BER Decay:** As SNR increases from 0 dB to 6 dB, the BER decreases exponentially from 7.56% to 0.17%. This steep drop illustrates the strong noise-suppression capability of antipodal binary signaling.

• **High-SNR Reliability (12 dB):** At 12 dB SNR ($\sigma^2 = 0.0315$), noise power is negligible. Zero bit errors occurred across 10,000 transmitted bits, demonstrating near-perfect reception for practical transmission.